Unit III: INTRODUCTION TO SQL (i)

Learning Outcomes

- Understand SQL and its importance
- Learn types of SQL commands
- Understand SQL Data Definition (DDL)
- Learn structure of SQL queries
- Use SQL operations with examples and outputs

What is SQL?

- SQL = Structured Query Language
- Used to interact with relational databases
- Data stored in tables (rows and columns)
- Used for:
  - Creating tables
  - Inserting data
  - Retrieving data
  - Updating data

Why is SQL Important?

- Standard database language
- Easy to use
- Handles large data efficiently
- Used in real-world systems

Types of SQL Commands

Type	Purpose	Example
DDL **Data Definition Language**	Define structure	CREATE
DML **Data Manipulation Language**	Modify data	INSERT
DQL **Data Query Language**	Retrieve data	SELECT
DCL **Data Control Language**	Control access	GRANT

Example: Student Table

StudentID	Name	Age
1	John	20
2	Alice	22
3	Bob	18
4	Anna	20
5	Mike	17

Example SQL Commands

- SELECT * FROM Student;
- Output:

StudentID	Name	Age
1	John	20
2	Alice	22
3	Bob	18
4	Anna	20
5	Mike	17

SQL Data Definition (DDL)

- Used to define structure
- Commands:
  - CREATE
  - ALTER
  - DROP

- Defines tables and constraints

CREATE TABLE

```
CREATE TABLE Student (
    StudentID INT PRIMARY KEY,
    Name VARCHAR(50),
    Age INT
);
```

- Creates a new table

Cont…

- ALTER TABLE

- ALTER TABLE Student ADD Email VARCHAR(100);

- Adds a Column

- DROP TABLE

- DROP TABLE Student;

- Deletes table permanently

Constraints

- PRIMARY KEY → Unique + Not Null
- FOREIGN KEY → Relationship
- NOT NULL → No empty values
- UNIQUE → No duplicates

Example with Constraints

```
CREATE TABLE Course (
    CourseID INT PRIMARY KEY,
    StudentID INT,
    FOREIGN KEY (StudentID) REFERENCES
Student(StudentID)
);
```

- Links two tables

Basic Structure of SQL Query

SELECT Name FROM Student WHERE Age > 18;

StudentID	Name	Age
1	John	20
2	Alice	22
3	Bob	18
4	Anna	20
5	Mike	17

Output:

Name
John
Alice
Anna

SELECT Statement

SELECT Name FROM Student;

Name
John
Alice
Bob
Anna
Mike

WHERE Clause

SELECT Name FROM Student WHERE Age > 18;

Name
John
Alice
Anna

ORDER BY

SELECT Name, Age FROM Student ORDER BY Age DESC;

Name	Age
Alice	22
John	20
Anna	20
Bob	18
Mike	17

GROUP BY

```
SELECT Age, COUNT(*)
FROM Student
GROUP BY Age;
```

Age	COUNT
17	1
18	1
20	2
22	1

HAVING

```
SELECT Age, COUNT(*)
FROM Student
GROUP BY Age
HAVING COUNT(*) > 1;
```

Age	COUNT
20	2

Additional Operations

- AND / OR / NOT
- LIKE
- BETWEEN
- IN

- Used for filtering data

AND Operator

```
SELECT * FROM Student
WHERE Age > 18 AND Name = 'John';
```

StudentID	Name	Age
1	John	20

LIKE Operator

```
SELECT * FROM Student
WHERE Name LIKE 'A%';
```

StudentID	Name	Age
2	Alice	22
4	Anna	20

BETWEEN Operator

```
SELECT * FROM Student
WHERE Age BETWEEN 18 AND 20;
```

StudentID	Name	Age
1	John	20
3	Bob	18
4	Anna	20

IN Operator

```
SELECT * FROM Student
WHERE Age BETWEEN 18 AND 20;
```

StudentID	Name	Age
2	Alice	22
3	Bob	18

Summary

- SQL is used to manage relational databases
- DDL defines structure
- SELECT retrieves data
- WHERE filters data
- Operators make queries powerful

THANK YOU, 1SWE!

References

Elmasri, R., & Navathe, S. B. (2016). *Fundamentals of database systems* (7th ed.). Pearson.

Halvorsen, H.-P. (2017). *Introduction to database systems*. Retrieved from https://www.halvorsen.blog